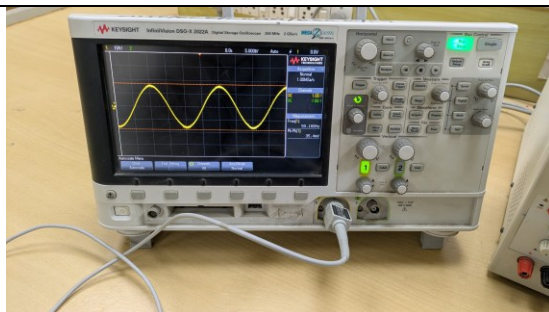


NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Electronics And Communication Engineering Department

Electronic Devices and Circuit Laboratory & Semiconductor Devices
Laboratory Equipment

Digital Storage Oscilloscope DSO

Displays and analyzes electrical waveforms. It captures, stores and digitizes signals for detailed time and frequency domain analysis. Features multiple channels, adjustable time base and advanced triggering options. Widely used for debugging circuits and studying transient behaviour of signals.



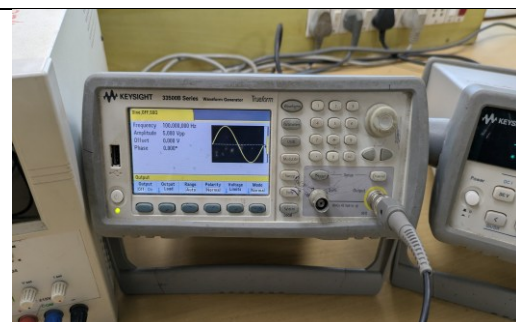
Keysight Digital Multimeter DMM

Measures voltage, current, resistance and capacitance. Offers high accuracy readings with auto-ranging and true RMS measurement capability. Also supports continuity, diode testing and frequency measurements. An essential instrument for circuit troubleshooting and component verification.



Keysight Function Generator

Produces test waveforms for electronic circuits. Generates sine, square, triangular and arbitrary waveforms over a wide frequency range. Amplitude, frequency and DC offset can be precisely adjusted as per requirement. Used for testing amplifiers, filters and other analog and digital circuits.



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Keithley Power Supply

Precision programmable DC power supply. Provides accurately controlled voltage and current outputs for sensitive experiments. Includes programmable limits and built-in protection against overload and short circuits. Ideal for device characterization and low-power semiconductor testing.



Power Supply (Scientific)

Stable regulated DC supply for laboratory use. Delivers constant voltage and current outputs with minimal ripple and noise. Front panel controls and meters allow easy adjustment and monitoring of the output. Commonly used to power breadboard circuits and student experiments.



Four Probe Resistivity Measurement

Measures the electrical resistivity of semiconductor materials. Uses four equally spaced probes to eliminate contact resistance errors. Enables study of resistivity variation with temperature and determination of energy band gap.



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Hall Effect Apparatus

Studies Hall effect and semiconductor properties. Determines carrier concentration, mobility and the type of majority charge carriers. Consists of an electromagnet, constant current source, gaussmeter and Hall probe. Helps students understand charge transport phenomena in semiconductor materials.

